

Stability by Design: Optimizing Prompt Properties in Small Language Models via Sensitivity Metrics NLP Final Project Proposal

Team: Sharon Levy, 315140152, sharonl4@mail.tau.ac.il; Yaniv Mualem, 209127612, yanivmualem@mail.tau.ac.il; Eden Daya, 315285759, edendaya@mail.tau.ac.il; Avner Fivelovich 314948811, avnerf@mail.tau.ac.il).

Moodle Group: 14

1. Project Description

Problem & motivation. Small language models (SLMs; $< 7\text{B}$ parameters) are attractive for efficient, privacy-preserving deployment, yet their outputs can be unstable: minor prompt edits (e.g., synonym swaps or added fillers) may change accuracy. We propose a quantitative approach for designing more stable prompts by linking prompt attributes to a sensitivity-based stability metric.

Related work. We combine: (1) a prompt-attribute taxonomy (Long et al., 2025), focusing on *metacognition* (self-verification), *structure* (output constraints), and *politeness* (conversational fillers); and (2) sensitivity metrics for prompting (Lu et al., 2023), which measure output variability under small input perturbations and report that higher sensitivity tends to accompany lower accuracy.

Research question & hypothesis. *Can optimizing prompt properties reduce input sensitivity in SLMs, and increase task performance?* We hypothesize **Structure** and **Metacognition** reduce sensitivity (toward $s \rightarrow 0$), while **Politeness** might provide limited benefit with no sensitivity increase or may increase sensitivity by adding irrelevant variation and reduce performance.

2. Methodology & Implementation Plan

Independent variables (prompt properties). We construct prompts isolating one property at a time:

- **Control:** standard zero-shot instruction.
- **+Metacognition:** add self-check triggers (e.g., “verify your answer”).
- **+Structure:** enforce an output schema (e.g., “Answer in: {Reasoning, Final}”).
- **+Politeness:** add fillers (e.g., “please”, “I would appreciate”).

Dependent variable (sensitivity). For each prompt, we will generate several perturbed variants (e.g., synonym replacements). Run the model on the original + variants and compute Variation Ratio $s = 1 - f_m / (N + 1)$, where f_m is the frequency of the modal output.

Tasks/datasets. We evaluate on reasoning-heavy benchmarks: **GSM8K** and **CSQA**.

Models. We start with small, accessible models (**Flan-T5-Base**, **Pythia-410M**) and scale up (**Flan-T5-Large**, **Pythia-1.4B**). If resources permit, we additionally test one larger model (e.g., **Phi-3-mini** or similar models) and report scaling trends.

3. Resources & Feasibility

Compute. We use the departmental Slurm cluster (**studentkillable**) for main runs and local CPU/GPU for pipeline development; we cap context length and batch sizes to avoid out-of-memory errors.

Implementation. Inference-only pipeline using Hugging Face **transformers** and **accelerate**; no full training is required.

4. Administration

If possible, we would like to present our proposal in class.

5. Bibliography

1. Long, J., et al. (2025). *What Makes a Good Natural Language Prompt? A Comprehensive Taxonomy of Prompt Attributes*. Proceedings of the ACL.
2. Lu, C., et al. (2023). *How are Prompts Different in Terms of Sensitivity? An Empirical Analysis of Input Stability*. Proceedings of the NAACL.

6. AI Usage Disclosure and Reflection

We used mainly GPT 5.2 to produce a LaTeX template (especially to fit everything into the 1-page limit) in Overleaf. Additionally, we used ChatGPT Deep Research mode and Gemini-3 Pro to come up with relevant papers that fit our research interest. Also, we used ChatGPT and Gemini-3 Pro to brainstorm the connection between prompt properties and sensitivity metrics. All scientific claims and experimental decisions were verified by the team.